# Supplementary material

Appendix A: Inclusion/Exclusion criteria.

| **Include** | **Exclude** |
|---|---|
| All quantitative empirical studies | Systematic reviews |
| Patients with Klinefelter syndrome, all ages | Meta-analysis |
| Outcome measures related to QoL and WHOQOL-100 dimensions (subsections) | Guidelines, position statements and case studies |
| Human studies only | Qualitative studies |
| Full text available | Animal studies |
| Peer-reviewed published studies | Reviews, editorials and opinion papers |
| Manuscripts in English | Duplicated cohorts |
| Studies with comparison groups | Studies without comparison groups |

Appendix B: Literature search strategy, CINAHL

Select / deselect all | Search with AND | Search with OR | Delete Searches | Refresh Search Results

| | Search ID# | Search Terms | Search Options | Actions |
|---|---|---|---|---|
| ☐ | S10 | S1 AND S9 | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (35) \| View Details \| Edit |
| ☐ | S9 | S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (643,721) \| View Details \| Edit |
| ☐ | S8 | spirituality OR religion OR personal beliefs | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (34,917) \| View Details \| Edit |
| ☐ | S7 | environment | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (184,599) \| View Details \| Edit |
| ☐ | S6 | social relations | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (4,340) \| View Details \| Edit |
| ☐ | S5 | level of independence | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (1,216) \| View Details \| Edit |
| ☐ | S4 | psychological | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (256,562) \| View Details \| Edit |
| ☐ | S3 | physical health | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (49,724) \| View Details \| Edit |
| ☐ | S2 | quality of life | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (191,559) \| View Details \| Edit |
| ☐ | S1 | klinefelter syndrome OR 48, xxyy syndrome OR 49 xxxxy syndrome OR xxxy males OR xxy syndrome OR xxy trisomy OR xxyy syndrome | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (472) \| View Details \| Edit |

# Appendix C: Literature search strategy, MEDLINE

Select / deselect all | Search with AND | Search with OR | Delete Searches | Refresh Search Results

| Search ID# | Search Terms | Search Options | Actions |
|---|---|---|---|
| S10 | S1 AND S9 | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (150) \| View Details \| Edit |
| S9 | S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 | **Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (1,592,497) \| View Details \| Edit |
| S8 | spirituality OR religion OR personal beliefs | **Limiters** - Date of Publication: -20200131; English Language<br>**Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (45,355) \| View Details \| Edit |
| S7 | environment | **Limiters** - Date of Publication: -20200131; English Language<br>**Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (744,220) \| View Details \| Edit |
| S6 | social relations | **Limiters** - Date of Publication: -20200131; English Language<br>**Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (7,028) \| View Details \| Edit |
| S5 | level of independence | **Limiters** - Date of Publication: -20200131; English Language<br>**Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (2,031) \| View Details \| Edit |
| S4 | psychological | **Limiters** - Date of Publication: -20200131; English Language<br>**Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (536,843) \| View Details \| Edit |
| S3 | physical health | **Limiters** - Date of Publication: -20200131; English Language<br>**Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (73,599) \| View Details \| Edit |
| S2 | quality of life | **Limiters** - Date of Publication: -20200131; English Language<br>**Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (309,792) \| View Details \| Edit |
| S1 | Klinefelter Syndrome OR 48, xxyy syndrome OR 49 xxxxy syndrome OR xxxy males OR xxy syndrome OR xxy trisomy | **Limiters** - Date of Publication: -20200131; English Language<br>**Expanders** - Apply equivalent subjects<br>**Search modes** - Boolean/Phrase | View Results (3,409) \| View Details \| Edit |

Appendix D: Joanna Briggs Quality appraisal tool: Analytical Cross-Sectional Studies.

| Author & year | Were the criteria for inclusion in the sample clearly defined? | Were the study subjects and the setting described in detail? | Was the exposure measured in a valid and reliable way? | Were objective, standard criteria used for measurement of the condition? | Were confounding factors identified? | Were strategies to deal with confounding factors stated? | Were the outcomes measured in a valid and reliable way? | Was appropriate statistical analysis used? |
|---|---|---|---|---|---|---|---|---|
| (Ferlin et al., 2018) | Y | Y | Y | Y | N | N | Y | Y |
| (Fisher et al., 2015) | Y | Y | Y | Y | Y | N | Y | Y |
| (Herlihy et al., 2011) | Y | Y | Y | Y | Y | N | Y | Y |
| (K,Sorensen, 1992) | Y | Y | Y | Y | N | N | Y | Y |
| (Rapp et al., 2018) | Y | Y | Y | Y | Y | Y | Y | Y |
| (Rijn, 2018) | Y | Y | Y | Y | N | N | Y | Y |
| (Skakkebaek et al., 2018) | Y | Y | Y | Y | Y | N | Y | Y |
| (Skakkebaek et al., 2017) | Y | Y | Y | Y | Unclear | N | Y | Y |
| (Van Rijn et al., 2014) | Y | Y | Y | Y | Y | N | Y | Y |
| (Van Rijn, Swaab, Aleman and Khan, 2008) | Y | Y | Y | Y | N | N | Y | Y |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| (Liberato et al., 2017) | Y | Y | Y | Y | N | N | Unclear | Y |
| (Fabrazzo et al.,2021) | Y | Y | Y | Y | Y | N | Y | Y |

Appendix E: Joanna Briggs Quality appraisal tool: Cohort Studies.

| Author & year | Were the two groups similar and recruited from the same population? | Were the exposures measured similarly to assign people to both exposed and unexposed groups? | Was the exposure measured in a valid and reliable way? | Were confounding factors identified? | Were strategies to deal with confounding factors stated? | Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)? | Were the outcomes measured in a valid and reliable way? | Was the follow up time reported and sufficient to be long enough for outcomes to occur? | Was follow up complete, and if not, were the reasons to loss to follow up described and explored? | Were strategies to address incomplete follow up utilized? | Was appropriate statistical analysis used? |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (Nielsen and Pelsen, 1987) | Y | N/A | Y | N | N | N/A | Y | Y | Y | N | Y |

Appendix F: Full text review. Included, Excluded studies with reasons

| # | Author & Year | Is a full text copy of the study available? | Are patients Males with a clinical KS diagnosis? | Is it primary research?<br><br>*If* **YES** *specify the type of study.* | Is the study measuring QOL?<br><br>*If* **YES** *which parameters does it meet e.g. 'Physical health, Environment, Psychological, Social Relations, Level of Independence'* | Can an explicit method or measurement of assessment of these parameters be identified?<br><br>*If* **YES**, *specify the measurement of assessment.* | If study is a cohort or an interventional study,<br><br>*What is the comparator?* | Is there an outcome associated with QOL and patients with KS?<br><br>*If* **YES**, *what is the outcome?* | Decision: Include or Exclude.<br><br>*If Excluded give reason.* |
|---|---|---|---|---|---|---|---|---|---|
| 1. | (Bender et al., 2001) | Y | N - Mix patient group of 23 patients with other genetic conditions, KS not labelled or clinical diagnosis confirmed. | Y - Cross-sectional study | Y - Psychological, Thinking, learning , memory. | Y – Wisconsin card sorting, Woodcock tests. WRAT, WAIS-R tests, WAIRS-R Full-scale IQ | Y - "Controls, including 11 male and 11 female siblings of propositi, were karyotypicaly normal" | Y - "In summary, the presence of an SCA results in increased potential for learning problems in childhood that persist into adulthood" | **Exclude**<br>-KS patients not defined or identified within results. |
| 2. | (Close, Fennoy, Smaldone and Reame, 2015) | Y | Y | Y – Cross-sectional study | Y – Psychological | Y –Questionnaires: Paediatric QOL inventory 4.0 health related- QOL, Piers-Harris 2 self concept scale, Coopersmith self-esteem inventory, Children's depression inventory. | N/A | Y -"Depending on the degree of phenotypic abnormality, boys with KS may be at risk for impaired QOL" | **Exclude**<br>-No comparator |
| 3. | (Ferlin et al., 2018) | Y | Y | Y – Part being assessed Cross-sectional study, second part Non-RCT | Y – Social relations, sexual activity. | Y - 15-Item International Index of Erectile Function, endocrine assessment, | Y - 60 age matched controls. | Y – "KS subjects have lower sexual desire, intercourse satisfaction" | **Include** |
| 4. | (Boratynska-Dabrowsk et al., 1970) | N | N/A | N/A | N/A | N/A | N/A | N/A | **Exclude**<br>– No full text |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 5. | (Herlihy, 2017) | N | N/A | N/A | N/A | N/A | N/A | N/A | **Exclude**<br>- No full text |
| 6. | (Close,2011) | Y | Y – 43 KS males | Y- Mixed methods (Cross-sectional and retrospective chart review) | Y - Physical health, Psychological | Y – Questionnaires, (Peds QL4.0), Coopersmith self-esteem inventory, Piers-Harris2 short form, (CDI), Physical problems frequencies. | N/A | Y – "Findings of the current study suggest that a high portion of KS (67.4%) have poor total QOL" | **Exclude**<br>-No comparator |
| 7. | (Fisher et al., 2015) | Y | Y | Y- Cross sectional study | Y – Social relations /sexual activity. Psychological/Thinking learning memory and concentration. | Y – Questionnaires. Psychometric tests. (AQ) (RME) (GIDYQ-AA) (SAST) (SCL-90-R) Wechsler Adult Intelligence scale-Revised. | Y- 43 healthy male controls. | Y- "KS is associated with hyper sexuality, paraphilic, and GB, which were mediated by obsessive compulsive and autistic traits." | **Include** |
| 8. | (Herlihy et al., 2011) | Y | Y – Males 18 years and older | Y – Cross sectional study | Y – Psychological, Environment, Social care. Bodily image, Self-esteem, general health, Sexual activity. | Y – (PWI), global life satisfaction, (MBSRQ-AS), Appearance Evaluation, Appearance Orientation subscale, (RSE), (K10), (SIS), (QRI), (Short form-1) | N/A | Y – Shows KS has a significant personal impact. | **Include** |
| 9. | (K,Sorensen, 1992) | Y | Y – 14 males with KS | Y- Cross sectional study | Y – Sexual activity, Psychological/ memory, learning and concentration. | Y – Wechsler Adult Intelligence Scale, School attainment. Mean scores for school difficulties including, Concentration, speech, interest. Experience of coitus and frequency of masturbation. | Y – 19 healthy male controls were used in the studies measuring QOL that have been identified. | Y – "subjects had increased height, reduced weight, impaired hearing, slightly lower intelligence, poor school performance, increased incidence of psychological consultation and lowered sexual activity. | **Include** |
| 10. | (Khan et al., 2018) | Y | Y – 154 KS diagnosed patients amongst mixed cohort. | Y – Cohort study | N | N | N | N | **Exclude**<br>-QOL is not explicitly measured. |

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| 11. | (Liberato et al., 2017) | Y | Y – 58 Adult male KS patients | Y – Cross sectional observation | Y – Psychological | Y – Clinical interview and questionnaires measuring. (SCID-ii), (MMPI-2), (SPM). | N/A | Y – PD prevalence was higher than in the general population. | **Include** |
| 12. | (Nielsen and Pelsen 1987) | Y | Y – 34 KS, 16 hypogonadal males without KS diagnosis. | Y - 20 year follow up study. | Y – Psychological, Sexual activity, Level of independence | Y – Anamnestic information and psychopathological symptoms. Occupation status & change in job responsibility, | Y – 16 hypogonadal males without KS diagnosis. | Y – Showed controls and KS patients had better than expected results in the tested measures, however this was post diagnosis and medical intervention for the previous twenty year. | **Include** |
| 13. | (Rapp et al., 2018) | Y | Y – 218 Klinefelter syndrome, other groups in study. KS clearly labelled throughout, where necessary. | Y- Multicentre cross-sectional clinical evaluation | Y – WHOQOL brief | Y – (WHOQOL-100 BREF) is used for KS patients. | Y – Healthy European populations. | Y – The study shows that all conditions included shows a that there's a significant lower QOL for the patients compared to controls. | **Include** |
| 14. | (Rijn, 2018) | Y | Y- 20 KS boys. | Y – Cross sectional study | Y – Psychological, Intellectual functioning and social anxiety. | Y – IQ, Social Anxiety Scale, | Y - 25, non-clinical controls between 8 and 19 years old. | Y – Mean level of intellectual functioning was significantly lower | **Include** |
| 15. | (Skakkebaek et al., 2014) | Y | Y – 73 KS males were used, Subgroups of Paternal, maternal, skewed and no skewed | Y – Cross sectional study | Y – Psychological, | Y – Autism spectrum Quotient scores. (SCL) | Y – 73 controls | N | **Exclude** -Unable to determine QOL outcomes from study as results cannot be analysed due to subgroup. |
| 16. | (Skakkebaek et al., 2018) | Y | Y – 132 KS males | Y – Cross-sectional study | Y – Psychological, Physical health, Social relations, Sexual activity. | Y – Questionnaires completed by KS patients. (IIEF-15), (WHOQOL-BREF). | Y – 313 matches controls | Y – KS status is associated with lower PQOL and MQOL. | **Include** |
| 17. | (Skakkebaek et al 2017) | Y | Y – 69 KS males | Y – Cross-sectional study | Y – Psychological, | Y – Questionnaires, IQ, (AQ scale), cognitive performance. | Y- 69 Controls | Y- KS is associated with lower intelligence and poorer social skills | **Include** |

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|---|---|---|---|---|---|---|---|---|---|
| 18. | (Skakkebaek et al., 2018) Anxiety and depression. | Y | N/A | N/A | N/A | N/A | N/A | N/A | **Exclude**<br>-Results in table one are identical to table one in, (Skakkebaek et al.,2017) The role of genes. |
| 19. | (Tartaglia et al., 2010) | Y | Y – 57 Children and adolescents with KS | Y – Cross sectional study | Y – Psychological, | Y – Questionnaires, Including (WISC-IV), (WASI), (SRS), ADHD symptoms, (ADI-R), (ADOS), | N/A | Y – Diagnosis of KS support having greater prevalence of psychological conditions and cognitive function. | **Exclude**<br>-No comparator |
| 20. | (Temple and Sanfilippo,2003) | Y | Y – 3 KS | Y – Case study | N/A | N/A | N/A | N/A | **Exclude**<br>-Case study |
| 21. | (Turriff, Levy and Biseecker, 2011) | Y | Y- 310 self reported XXY aged 14-75 | Y – Cross sectional study | Y – Psychological | Y – Questionnaire using online websites. (CES-D), | N/A | Y – KS may be at increased risk for depressive symptoms. | **Exclude**<br>-No comparator |
| 22. | (Turriff, Levy and Biesecker,2015) | Y | Y- 310 self reported XXY aged 14-75 | N/A | N/A | N/A | N/A | N/A | **Exclude**<br>-Participation cohort seemingly used for (Turriff, Levy and Biseecker,2011) |
| 23. | (Turriff, Macnamara, Levy and Bisecker, 2016) | Y | Y- 310 self reported XXY aged 14-75 | Y – Thematic analysis. | N/A | N/A | N/A | N/A | **Exclude**<br>-Thematic analysis |
| 24. | (Van Rijn and Swaab, 2011) | Y | Y – 73 Ks subjects | Y – Cross sectional study. | Y – Psychological, | Y- Wechsler Intelligence scales, Autism questionnaire, Schizotypal personality Questionnaire. | Y – 93 age matched controls | Y – Psychopathology traits were significantly higher with KS than the control group. | **Exclude**<br>-Patient group recruited from same pool as (Rijn et al., 2008 & Rijn et al 2006). |

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|---|---|---|---|---|---|---|---|---|---|
| 25. | (Van Rijn and Swaab,2015) | Y | Y – 23 KS boys, clearly labelled, other participants 17 girls with Trisomy X, XXX. | Y – Cross sectional study | Y – Psychological | N– Outcome measures not shown for KS males | Y – 100 non clinical controls, 47 boys and 53 girls. | Y – Executive function is impacted from an additional chromosome affecting everyday function. | **Exclude**<br>-outcome measures |
| 26. | (Van Rijn et al., 2014) | Y | Y – 34 boys with KS and 26 girls with Trisomy X | Y – Cross sectional study | Y- Psychological, social relations | Y – IQ, (SSRS), ( SRS), social anxiety. | Y – 106 non clinical controls 46 boys and 60 girls. | Y – Score between boys and girls were similar with an extra X, scores showed more prevalence in the test than the controls. Differences between boys and girls shown, in results. | **Include** |
| 27. | (Van Rijn et al., 2016) | Y | Y – 29 KS boys and 21 Trisomy X girls. Various results explain both girls and boys results. | Y – Cross sectional study | Y – Psychological, thinking, learning and memory. | Y – IQ, Cognitive performance scores, | Y – 103 non-clinical controls. | Y – Intellectual functioning was lower in the KS group than controls. Early life stress has a greater negative on those with an extra chromosome than those without. | **Exclude**<br>-Same patient group used as (Van Rijn et al., 2013) |
| 28. | (Van Rijn, de Sonneville and Swaab, 2018) | Y | Y- 70 males with KS. | Y – Cross sectional study | Y – Psychological, thinking, learning and memory. Social relations | Y – Wechsler Intelligence scales, Full scale IQ:FSIQ), (ANT), | N/A | Y – KS is linked to lower social cognition with regards to ToM, and fits other studies measuring similar outcomes. | **Exclude**<br>-No comparator |
| 29. | (Van Rijn, Swaab, Aleman and Khan, 2006) | Y | Y – 32 KS men | Y – Cross sectional study | Y – Psychological, Social relations | Y – (Raven's Advanced Progressive metrics, short form), (NART), Benton and Van Allen Test, | Y - 26 men from the general population. | Y – KS had greator social cognition disturbances than that of the general population. | **Exclude**<br>-Similar patient group from same participant pool as,(Van Rijn, Swaab, Aleman and Khan,2008). |
| 30. | (Van Rijn, Swaab, Aleman and Khan, 2008) | Y | Y – 31 KS men | Y – Cross sectional study | Y – Psychological, Social relations | Y – (Ravens Advanced Progressive metrics, (NART), scale for interpersonal behaviour(SIB), (Autism spectrum Quotient), | Y – two groups of 24 and 20 male controls. | Y – KS men had increased distress in social situations and less engagement in certain social situations. Ks also had significant higher autism traits. | **Include** |

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|---|---|---|---|---|---|---|---|---|---|
| 31 | (Fabrazzo et al., 2021) | Y | Y – 23 Men with KS post 1-year androgen therapy. | Y – Cross-sectional study | Y – Overal and psychological measures | Y – Q-LES-Q, MMSE, SCL—90-R, TCI-R | Y – 23 healthy controls | Y – KS patients, treated for one year with testosterone undecanoate, do not yield a QoL compared to matched healthy controls. | **Include** |